



The Scots College

Year 12 Mathematics Extension 2

Assessment 4

August 2005

GENERAL INSTRUCTIONS

- | | |
|---|---|
| <ul style="list-style-type: none">▪ Working time - 3 hours▪ Write using blue or black pen▪ Board approved calculators may be used▪ All necessary working should be shown in every question▪ Standard Integrals Table attached | <p>TOTAL MARKS: 120</p> <p>WEIGHTING: 40 %</p> <ul style="list-style-type: none">▪ Start each QUESTION on a new answer booklet |
|---|---|

QUESTION 1 (15 marks)

1

a) Evaluate $|3 + 2i|$

2

b) i) If $v = \frac{1+i\sqrt{3}}{2}$ show that $v^3 = -1$.

2

ii) Hence calculate v^{10} .

c) If z is a complex number so that $|z| = 2$ and $\arg z = \frac{\pi}{6}$, mark clearly on the same Argand diagram the points representing the complex numbers:

i) z ii) iz iii) $\bar{z}$ iv) $\frac{1}{z}$ v) $z\bar{z}$ vi) z^2 vii) $z^2 + z$ viii) $z^2 - z$ **10**

QUESTION 2 (15 marks)

a) Find $\int \frac{dx}{x^2 - 6x + 13}$

2

b) Find $\int \tan x \sec^2 x \, dx$

2

c) i) Show that $f(x) = \sin^{-1}x$ is an odd function.

2

ii) Hence or otherwise find $\int_{-1}^1 (\sin^{-1} x)^3 dx$

1

d) $\int_0^{\sqrt{2}} \sqrt{4-x^2} \, dx$

4

e) $\int e^x \cos x \, dx$

4

QUESTION 3 (10 marks)

a) Use the method of cylindrical shells to find the volume of the solid (paraboloid) obtained when the region between the curve $y = \frac{1}{2}\sqrt{x-2}$, the x-axis and the line $x = 6$ is rotated about the x axis. 4

b) Prove $\frac{d^2x}{dt^2} = \frac{d}{dx}\left(\frac{1}{2}v^2\right)$, where x denotes displacement, and v denotes velocity. 2

c) The acceleration of a particle moving in a straight line is given by $\ddot{x} = xe^x$, where x is the displacement from 0. The particle is initially at rest.

The particle starts at $x = 0$.

i) Prove that $v^2 = 2e^x(x-1) + 2$ 3

ii) Describe the subsequent motion of the particle after it leaves the origin and explain why the particle can only move in one direction 1

QUESTION 4 (18 marks)

a) The equation $x^3 - x^2 - 3x + 2 = 0$ has roots α, β, γ . Find the monic polynomial equation with roots $\alpha^2, \beta^2, \gamma^2$. 4

b) If $x = \alpha$ is a double root of the equation $P(x) = 0$, show that $x = \alpha$ is a root of the equation $P'(x) = 0$. 4

c) i) Show that $1 + i$ is a root of the polynomial $Q(x) = x^3 + x^2 - 4x + 6$ 2

ii) hence resolve $Q(x)$ into irreducible factors over the complex number field. 3

d) If α, β, γ are the roots of the cubic equation $x^3 + qx + r = 0$, prove that $(\beta - \gamma)^2 + (\gamma - \alpha)^2 + (\alpha - \beta)^2 = -6q$. 5

QUESTION 5 (18 marks)

The ellipse $\mathcal{E}$ has the cartesian equation $\frac{x^2}{25} + \frac{y^2}{16} = 1$.

- i) Write down the eccentricity 1
- ii) Write down the coordinates of the foci S and S' 1
- iii) Write down the equations of the directrices. 1
- iv) Sketch the ellipse $\mathcal{E}$. 1
- v) Show that any point P on $\mathcal{E}$ can be represented by the coordinates $(5\cos\theta, 4\sin\theta)$ 1
- vi) Prove that $PS + PS'$ is independent of the position of P on the ellipse $\mathcal{E}$. 3
- vii) Show that the equation of the normal N at the point P on the ellipse $\mathcal{E}$ is 2
 $5\sin\theta x - 4\cos\theta y = 9\sin\theta\cos\theta$
- viii) If this normal meets the major axis of the ellipse in M and the minor axis in N , 3
prove that $\frac{PM}{PN} = \frac{16}{25}$.
- ix) Also show that the line PN bisects the angle $S'PS$. 5

QUESTION 6 (14 marks)

- i) By considering the curve $g(x) = x^6 - 4x^5 + 4x^4$, sketch the graph of 4
 $f(x) = x^6 - 4x^5 + 4x^4 - 1$ showing that it has 4 real zeroes.
- On different diagrams sketch the curves:
- ii) $y = |f(x)|$ 2
- iii) $y = f(|x|)$ 2
- iv) $y^2 = f(x)$ 3
- v) Calculate the slope of the curve $y^2 = f(x)$ at any point x and describe the nature of the curve at a zero of $f(x)$. 3

QUESTION 7 (15 marks)

a) A parachutist of M kilograms is dropped from a stationary helicopter of height H metres above the ground. The parachutist experiences air resistance during its fall equal to MkV^2 , where V is its velocity in metres per second and k is a positive constant. Let x be the distance in metres of the parachutist from the helicopter, measured positively as it falls.

i) Show that the equation of motion of the parachutist is $\ddot{x} = g - kV^2$, where g is the acceleration due to gravity. 1

ii) Find V^2 as a function of x . 4

iii) Find the velocity U of the parachutist as he hits the ground in terms of g , k and H . 1

iv) Find the velocity of the parachutist as he hits the ground if air resistance is neglected. 2

b)

i) Prove the identity $\cos 3A = 4\cos^3 A - 3\cos A$ 2

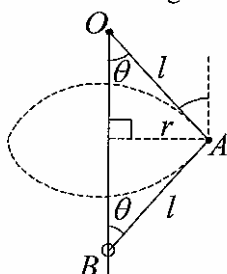
ii) Show that $x = 2\sqrt{2} \cos A$ is a root of the equation $x^3 - 6x + 2 = 0$ provided that $\cos 3A = -\frac{1}{2\sqrt{2}}$ 2

iii) Find the three roots of the equation $x^3 - 6x + 2 = 0$, using the results from part (ii) above. Give your answer to three decimal places. 3

QUESTION 8 (15 marks)

a) A particle **A** of mass $2m$ is attached by a light inextensible string of length l to a fixed point **O** and is also attached by another light inextensible string of the same length to a small ring **B** of mass $3m$ which can slide on a fixed smooth vertical wire passing through **O**. The particle **A** describes a horizontal circle of radius r , and **OA** is inclined at an angle $\theta = \frac{\pi}{3}$ with the downward vertical.

Dimension diagram



$$\theta = \frac{\pi}{3}$$

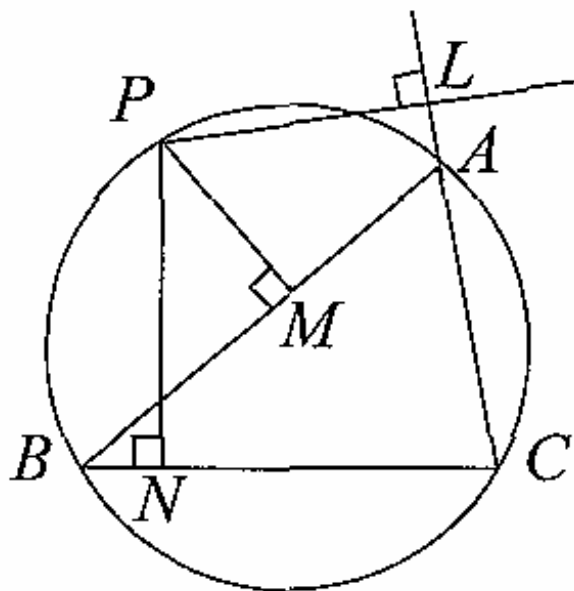
i) Find the tension in the strings **OA** and **AB** 5

ii) Find the angular velocity of **A**. 3

iii) Describe what happens to the system as the angular velocity increases. 1

b) $\triangle ABC$ is a triangle inscribed in the circle. P is a point on the minor arc AB . The points L , M , and N are the feet of the perpendiculars from P to CA produced, AB , and BC respectively.

Copy the diagram into your answer booklet and show that L , M and N are collinear.



END OF EXAM

Ex 2: 2005 Trial Solutions

Question 1

$$a) \sqrt{3^2 + 2^2} = \sqrt{13} \quad \checkmark \quad (1)$$

$$b) v = \frac{1 + i\sqrt{3}}{2}$$

$$v = \cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \quad \checkmark$$

$$v^3 = \cos \pi + i \sin \pi \quad \checkmark \quad (2)$$

$$v^3 = -1 + 0i$$

$$v^3 = -1$$

$$ii) v^{10} = (v^3)^3 v \quad \checkmark$$

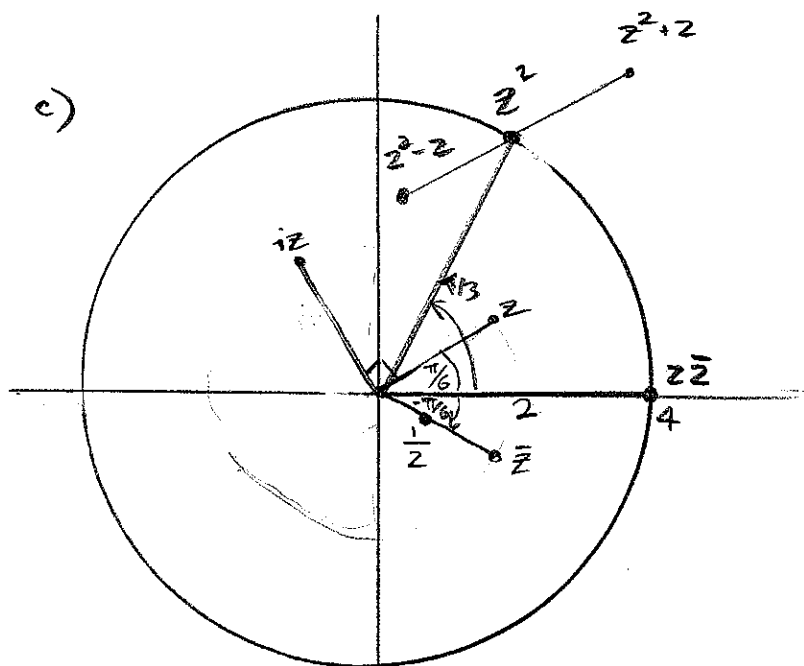
$$= (-1)^3 v \quad (2)$$

$$= -v$$

$$= -\frac{1}{2} - \frac{i\sqrt{3}}{2} \quad \checkmark$$

$$c) vii) z^2 + z \text{ using vector addition} \quad \checkmark \checkmark$$

$$viii) z^2 - z \text{ using vector subtraction} \quad \checkmark \checkmark$$



$$ii) \text{ multiply by } 90^\circ \quad \checkmark$$

$$iii) z = 2(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6})$$

$$\therefore iz = 2(i \cos \frac{\pi}{6} - \sin \frac{\pi}{6})$$

$$= 2\left(i \frac{\sqrt{3}}{2} - \frac{1}{2}\right)$$

$$= -1 + i\sqrt{3} \quad \checkmark$$

$$iii) \bar{z} = 2(\cos \frac{\pi}{6} - i \sin \frac{\pi}{6})$$

the conjugate of z is the reflection of pt. z in the x axis. $\checkmark$

$$iv) \frac{1}{z} = \frac{1}{2(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6})}$$

$$z^{-1} = \frac{1}{2}(\cos \frac{\pi}{6} - i \sin \frac{\pi}{6}) \quad \checkmark$$

$$\therefore \frac{1}{2} \text{ has modulus} = \frac{1}{2}$$

$$\text{argument} = -\frac{\pi}{6}$$

$$v) z\bar{z} = |r|^2 = 4 \quad \checkmark$$

$$vi) (z)^2 = 2^2(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3}) \quad \checkmark$$

by de Moivre's Theorem.

$$= 4(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3})$$

Question 2

a) $\int \frac{dx}{x^2 - 6x + 13}$

$= \int \frac{1}{(x-3)^2 + 4} dx$ ✓

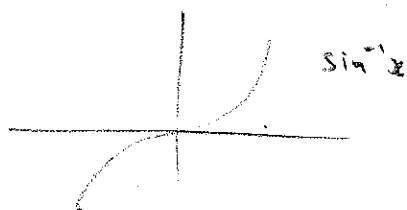
$= \frac{1}{2} \tan^{-1} \left(\frac{x-3}{2} \right) + C$ ✓ (2)

b) $\int \tan x \sec^2 x dx$

$= \frac{1}{2} (\tan x)^2 + C$ ✓ (1)

$= \frac{1}{2} \tan^2 x + C$

c) $f(-x) = -f(x)$
for odd functions



$f(x) = \sin^{-1}(x)$

$f(1) = \pi$

$f(-1) = -\pi$

as $f(-x) = -f(x)$ ✓

$f(x) = \sin^{-1}(x)$ is an odd function

ii) from i) $\sin^{-1} x$ is odd ✓
 $\therefore (\sin^{-1} x)^3$ is also odd

$\therefore \int_{-1}^1 (\sin^{-1} x)^3 dx = 0$ due to symmetry.

d) $\int_0^{\sqrt{2}} \sqrt{4-x^2} dx$ ✓✓

$= \int_0^{\pi/4} \sqrt{4-4\sin^2 \theta} \cdot 2\cos \theta d\theta$
($x = 2\sin \theta$)

$\frac{dx}{d\theta} = 2\cos \theta$

$= \int_0^{\pi/4} 2\cos \theta \cdot 2\cos \theta d\theta$ ✓

$= \int_0^{\pi/4} 4\cos^2 \theta d\theta$

$= \int_0^{\pi/4} 4 \left[\frac{1}{2} (1 + \cos 2\theta) \right] d\theta$ ✓

$= \int_0^{\pi/4} 2 + 2\cos 2\theta d\theta$

$= [2\theta + \sin 2\theta]_0^{\pi/4}$

$= \frac{\pi}{2} + 1$ ✓ (4)

e) $\int e^x \cos x dx$

let
 $v' = e^x$ $u = \cos x$
 $v = e^x$ $u' = -\sin x$

$= [e^x \cos x] - \int -\sin x (e^x) dx$ ✓

$= e^x \cos x + \int \sin x (e^x) dx$ ✓

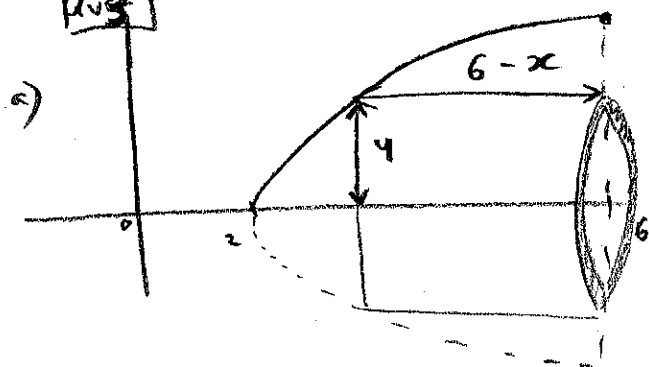
$= e^x \cos x + e^x \sin x - \int e^x \cos x$

$\therefore 2 \int e^x \cos x dx = e^x \cos x + e^x \sin x$ ✓ (4)

$\int e^x \cos x dx = \frac{1}{2} e^x (\cos x + \sin x) + C$

SECTION B

Qv3



$$y = \frac{1}{2}\sqrt{x-2}, \quad 2y = \sqrt{x-2}$$

$$r = y$$

$$h = 6 - x$$

$$4y^2 = x - 2$$

$$x = 4y^2 + 2$$

$$h = 6 - (4y^2 + 2) = 4 - 4y^2$$

Vol. Shell

$$\delta V = 2\pi r h \delta y$$

$$= 2\pi y \times (4 - 4y^2) \delta y$$

$$= 8\pi y (1 - y^2) \delta y$$

Volume paraboloid

$$V = \int_0^1 8\pi y (1 - y^2) \delta y$$

$$= 8\pi \int_0^1 y - y^3 \delta y$$

$$= 8\pi \left[\frac{y^2}{2} - \frac{y^4}{4} \right]_0^1$$

$$= 2\pi$$

$\therefore$ Volume of Paraboloid is 2π cubic units.

$$b) \frac{d^2x}{dt^2} = \frac{dv}{dt}$$

$$= \frac{dv}{dx} \cdot \frac{dx}{dt}$$

$$= v \cdot \frac{dv}{dx}$$

and

$$\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = \frac{d}{dv} \left(\frac{1}{2} v^2 \right) \times \frac{dv}{dx}$$

$$= v \cdot \frac{dv}{dx}$$

$$i) \frac{1}{2} v^2 = \int x e^x dx$$

$$= x e^x - \int e^x dx$$

$$= x e^x - e^x + C$$

$$v^2 = 2x e^x - 2e^x + C$$

$$0 = -2 + C$$

$$\therefore C = 2$$

$$v^2 = 2x e^x - 2e^x + 2$$

$$v^2 = 2e^x(x-1) + 2$$

$$ii) v^2 \geq 0$$

$$x < 0, v^2 < 0$$

$\therefore x$ must remain +ve.

as $x \uparrow, v \uparrow$

$$\frac{d^2x}{dt^2} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$$

(2)

Q4

a) $\alpha^2, \beta^2, \gamma^2$ each satisfy

$$(x^{\frac{1}{2}})^3 - (x^{\frac{1}{2}})^2 - 3(x^{\frac{1}{2}}) + 2 = 0 \quad \checkmark$$

$$x^{\frac{3}{2}} - 3x^{\frac{1}{2}} = x - 2 \quad \checkmark$$

$$x^3 - 6x^2 + 9x = x^2 - 4x + 4 \quad \checkmark$$

$$x^3 - 7x^2 + 13x - 4 = 0 \quad \checkmark$$

b) $P(x) = 0$ has a double root $x = \alpha$

$$\therefore P(x) = (x - \alpha)^2 \cdot Q(x) \quad \checkmark$$

$$\begin{aligned} \therefore P'(x) &= (x - \alpha)^2 \cdot Q'(x) + Q(x) \cdot 2(x - \alpha) \\ &= (x - \alpha) \left[(x - \alpha) \cdot Q'(x) + 2 \cdot Q(x) \right] \end{aligned}$$

$$\therefore P'(\alpha) = 0 \quad \checkmark$$

$\therefore P'(x) = 0$ has a root at $x = \alpha$.

$$c) (1+i)^2 = 1 + 2i + i^2 = 2i \quad \checkmark$$

$$(1+i)^3 = 2i(1+i) = 2i - 2 \quad \checkmark$$

$$P(1+i) = (1+i)^3 + (1+i)^2 - 4(1+i) + 6$$

$$= 2i - 2 + 2i - 4 - 4i + 6 \quad \checkmark$$

$$= 0$$

hence $(1+i)$ is a root of $P(x)$

ii) as $(1+i)$ is a root of polynomial $P(x)$, the complex roots occur in conjugate pairs.
thus $1-i$ is also a root of $P(x)$.

$P(x)$ has roots $1+i, 1-i, \gamma$

where γ is the 3rd root.

sum of roots = -1

$$\therefore 1+i + 1-i + \gamma = -1 \quad \checkmark$$

$$\gamma = -3$$

ii) cont...

factors of $P(x)$ are:

$$\{x - (1+i)\} \{x - (1-i)\} (x+3) \quad \checkmark$$

$$\begin{aligned} d) (\beta - \gamma)^2 + (\gamma - \alpha)^2 + (\alpha - \beta)^2 \\ = 2(\alpha^2 + \beta^2 + \gamma^2) - 2(\alpha\beta + \beta\gamma + \gamma\alpha) \\ = 2\{(\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \gamma\alpha)\} \\ - 2(\alpha\beta + \beta\gamma + \gamma\alpha) \end{aligned}$$

Now

$$\sum \alpha = 0$$

$$\sum \alpha\beta = q$$

$$\sum \alpha\beta\gamma = -r$$

$$= 2\{0 - 2q\} - 2q$$

$$= -6q$$

Qv 5 $\frac{x^2}{25} + \frac{y^2}{16} = 1$

$\therefore a = 5, b = 4$

i) from $b^2 = a^2(1 - e^2)$

$16 = 25(1 - e^2)$

$\therefore e^2 = 1 - \frac{16}{25}$

$e = \frac{3}{5}$

ii) foci $(\pm ae, 0)$

$(\pm 5 \cdot \frac{3}{5}, 0)$

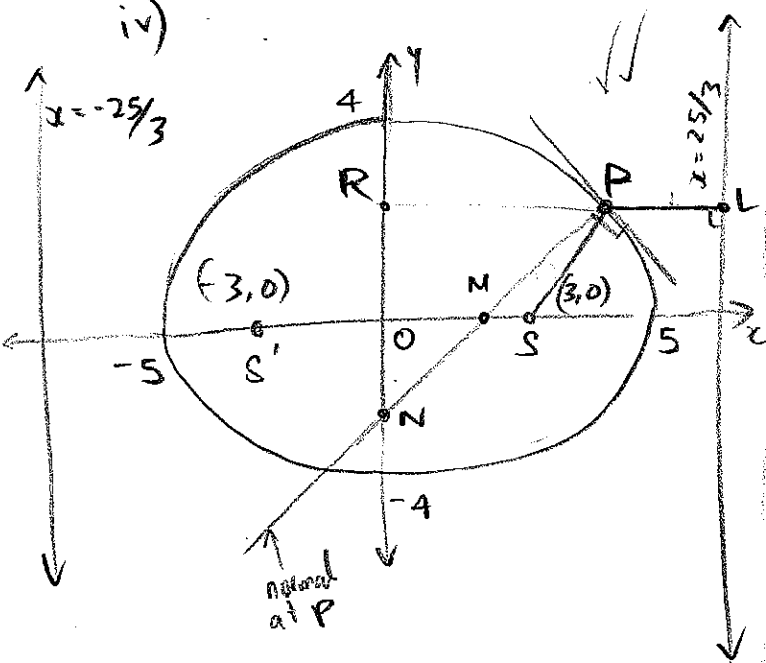
$S(3, 0) \quad S'(-3, 0)$

iii) directrices $x = \pm a/e$

$x = \pm 5 / (\frac{3}{5})$

$x = \pm 25/3$

iv)



v) Sub $P(5\cos\theta, 4\sin\theta)$ into eqn. E

$\frac{x^2}{25} + \frac{y^2}{16} = \frac{(5\cos\theta)^2}{25} + \frac{(4\sin\theta)^2}{16}$

$= \cos^2\theta + \sin^2\theta = 1, \therefore P \text{ lies on } E.$

vi) Definition of ellipse

$e = \frac{\text{dist. } P \text{ to } S}{\text{dis. } P \text{ to directrix}}$

$e = \frac{PS}{PL}$

$PS = e \cdot PL$

$= \frac{3}{5} \left(\frac{25}{3} - 5\cos\theta \right)$

$PS = 5 - 3\cos\theta$

$\frac{PS'}{PL'} = e$

$PS' = e \cdot PL'$

$= \frac{3}{5} \left(\frac{25}{3} + 5\cos\theta \right)$

$PS' = 5 + 3\cos\theta$

$\therefore PS + PS' = 10$, which is independent of P on curve E

vii) let $x = 5\cos\theta$

$y = 4\sin\theta$

$\frac{dx}{d\theta} = -5\sin\theta$

$\frac{dy}{d\theta} = 4\cos\theta$

$\frac{dy}{dx} = \frac{4\cos\theta}{-5\sin\theta}$

equation of tangent

gradient of normal

$= \frac{5\sin\theta}{4\cos\theta}$

eqn normal

$y - 4\sin\theta = \frac{5\sin\theta}{4\cos\theta} (x - 5\cos\theta)$

$\therefore 5\sin\theta x - 4\cos\theta y = 9\sin\theta\cos\theta$
is the eqn of normal.

viii) normal at P meets major axis when $y=0$

$$5\sin\theta x = 9\sin\theta\cos\theta$$

$$x = \frac{9\cos\theta}{5}$$

$$M\left(\frac{9\cos\theta}{5}, 0\right)$$

Normal meets minor axis when $x=0$

$$-4\cos\theta y = 9\sin\theta\cos\theta$$

$$y = -\frac{9\sin\theta}{4}$$

$$N\left(0, -\frac{9\sin\theta}{4}\right)$$

Let R be point on y axis from the extension of LP

$\triangle OMN \parallel \triangle RPN$ (AAA)

$$\therefore \frac{NM}{NP} = \frac{NO}{NR} = \frac{OM}{RP}$$

$$\frac{NM}{NP} = \frac{9\cos\theta/5}{5\cos\theta} = \frac{9}{25}$$

(taking horizontal distances)

$$\text{Thus } \frac{PM}{PN} = \frac{25-9}{25} = \frac{16}{25}$$

or use distance formula.

$\therefore \hat{MPS} = \hat{S'PM}$
hence normal at P bisects $\angle S'PS$.

ix) Let $\sin\theta = s$
 $\cos\theta = c$

$$\text{Grad PS} = \frac{4\sin\theta}{5\cos\theta - 3} = \frac{4s}{5c-3}$$

$$\text{Grad PN} = \frac{9\sin\theta}{4} \times \frac{5}{9\cos\theta} = \frac{5s}{4c}$$

$$\text{Grad PS}' = \frac{4\sin\theta}{5\cos\theta + 3} = \frac{4s}{5c+3}$$

$$\tan \hat{MPS} = \left| \frac{\frac{5s}{4c} - \frac{4s}{5c-3}}{1 + \frac{5s}{4c} \cdot \frac{4s}{5c-3}} \right|$$

$$= \left| \frac{25sc - 15s - 16sc}{20c^2 - 12c + 20s^2} \right|$$

$$= \left| \frac{9sc - 15s}{20 - 12c} \right| = \left| \frac{3s(3c-5)}{4(5-3c)} \right|$$

$$= \frac{3s}{4}$$

$$= \frac{3\sin\theta}{4}$$

$$\tan \hat{S'PM} = \left| \frac{\frac{5s}{4c} - \frac{4s}{5c+3}}{1 + \frac{5s}{4c} \cdot \frac{4s}{5c+3}} \right|$$

$$= \frac{25sc + 15s - 16sc}{20c^2 + 12c + 20s^2}$$

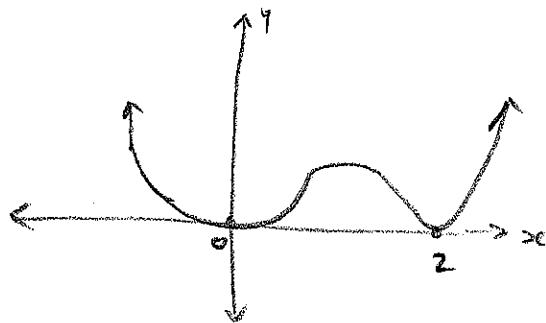
$$= \left| \frac{9sc + 15s}{20 + 12c} \right| = \left| \frac{3s(3c+5)}{4(5+3c)} \right|$$

$$= \frac{3\sin\theta}{4}$$

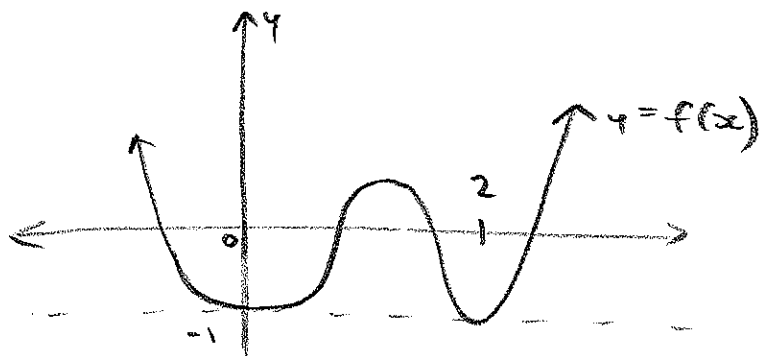
Q6 $f(x) = x^6 - 4x^5 + 4x^4 - 1$

a) consider
 $y = x^6 - 4x^5 + 4x^4$
 $y = x^4(x^2 - 4x + 4)$
 $y = x^4(x-2)^2$

$\therefore$ zeroes at $x=0$ and $x=2$



Shift curve down 1 unit for $f(x) = x^6 - 4x^5 + 4x^4 - 1$



$\therefore$ two turning points are $(0, -1)$ and $(2, -1)$.

Check for max. + p b/w $x=0$ + $x=2$ and above x axis

$$f'(x) = 6x^5 - 20x^4 + 16x^3$$

$$= 2x^3(3x^2 - 10x + 8)$$

$$= 2x^3(x-2)(ax+b)$$

as $x=2$ is a solⁿ to $f'(x)$

$$= 2x^3(x-2)(3x-4)$$

as double root occurs here

by equating coefficients

when $f'(x) = 0$

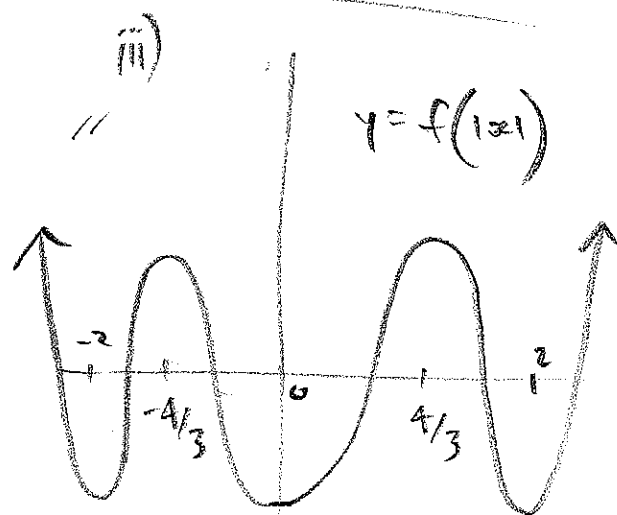
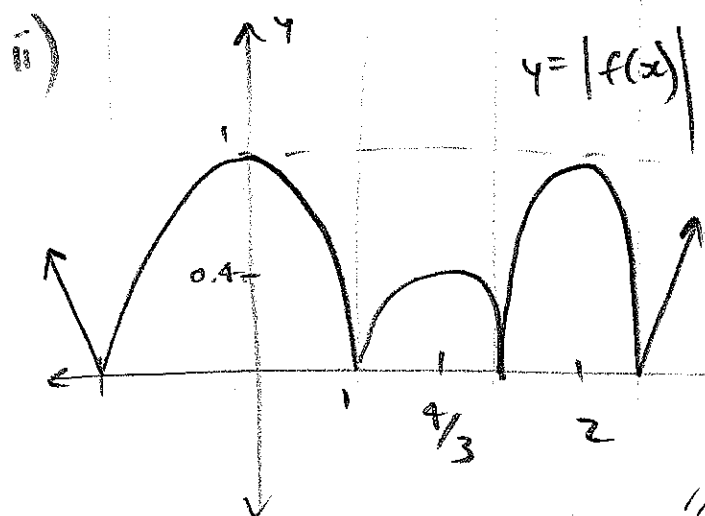
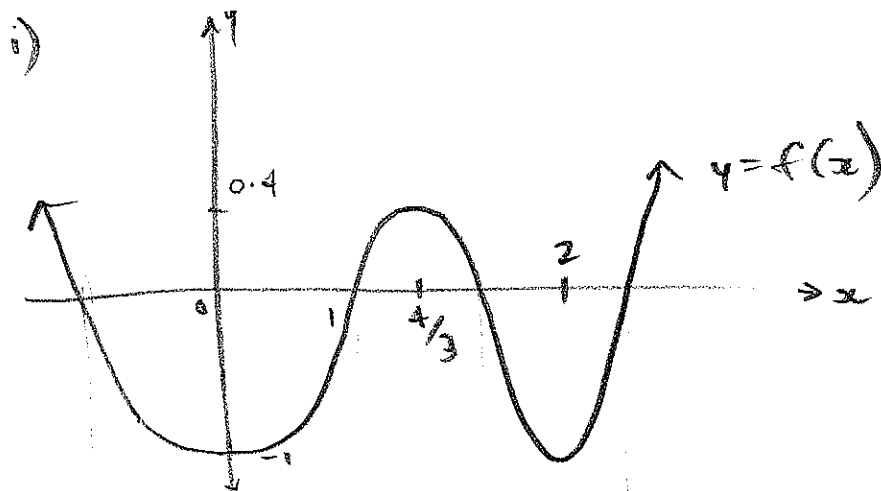
$$x=0, x=2, x = \frac{4}{3}$$

Sub. $x = \frac{4}{3}$ into $y = f(x)$

$y \approx 0.4$ $\therefore$ max t.p above y axis

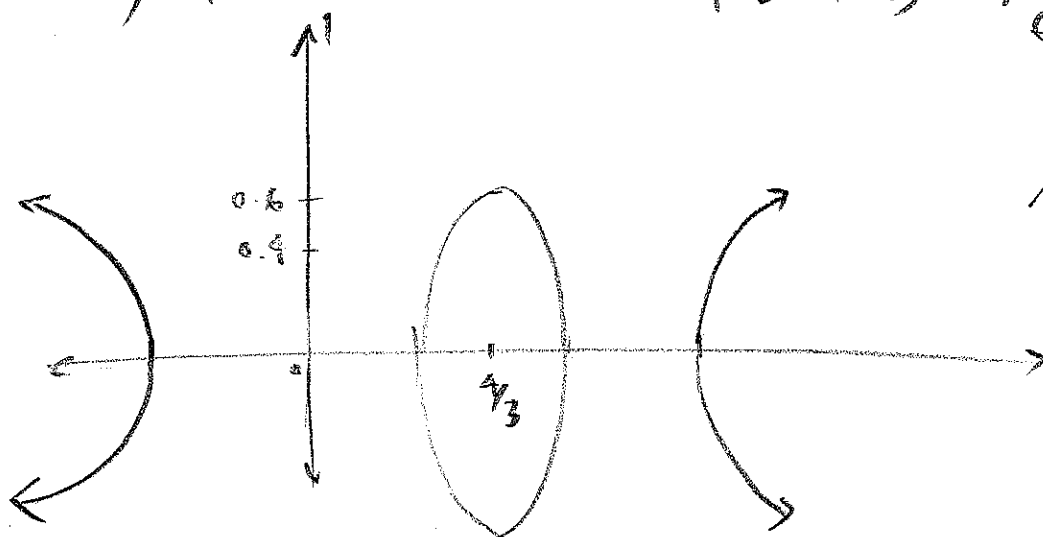
creating 4 roots.

or find 4 roots of $y = f(x)$ then sketch curve.



iv) $y^2 = f(x)$

$\therefore y = \pm \sqrt{f(x)}$, defined for $f(x) \geq 0$



This is undefined for $f(x) = 0$
 $\therefore$ at zeroes of $f(x)$ the curve has vertical tangents.

iv) $y^2 = f(x)$

$$\frac{d}{dy} y^2 \cdot \frac{dy}{dx} = \frac{d}{dx} f(x)$$

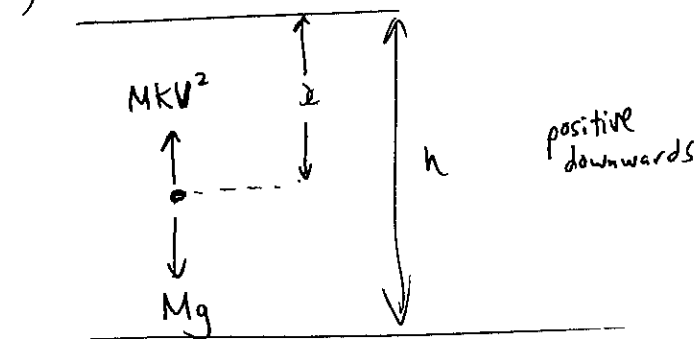
$$2y \cdot \frac{dy}{dx} = f'(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{2y}$$

$$= \pm \frac{f'(x)}{2\sqrt{f(x)}}$$

Q.7

i)



Equation of Motion

$$M\ddot{x} = Mg - MkV^2$$

$$\ddot{x} = g - kV^2$$

iii) when $x=h$, $v=V$

$$V^2 = \frac{g}{k} (1 - e^{-2kh})$$

$$V = \sqrt{\frac{g}{k} (1 - e^{-2kh})}$$

iv) Without air resistance, the equation of motion is

$$\ddot{x} = g$$

$$v \frac{dv}{dx} = g$$

$$\frac{1}{2}v^2 = gx + C$$

when $x=0$, $v=0$, $C=0$

$$\therefore v^2 = 2gx$$

when $x=h$, $v=V$

$$\text{so } V^2 = 2gh$$

$$V = \sqrt{2gh}$$

$$\text{ii) } v \frac{dv}{dx} = g - kV^2$$

$$\frac{v}{g - kV^2} dv = dx$$

$$\int \frac{v dv}{g - kV^2} = \int dx$$

$$-\frac{1}{2k} \log_e(g - kV^2) = x + C$$

when $x=0$, $v=0$

$$\text{so } C = -\frac{1}{2k} \log_e g$$

$$-\frac{1}{2k} \log_e(g - kV^2) = x - \frac{1}{2k} \log_e g$$

$$x = \frac{1}{2k} \log_e \left(\frac{g}{g - kV^2} \right)$$

$$2kx = \log_e \left(\frac{g}{g - kV^2} \right)$$

$$e^{2kx} = \frac{g}{g - kV^2}$$

$$\frac{g - kV^2}{g} = e^{-2kx}$$

$$g - kV^2 = g e^{-2kx}$$

$$kV^2 = g(1 - e^{-2kx})$$

$$V^2 = \frac{g}{k} (1 - e^{-2kx})$$

7b

a) i) $\cos 3A = \cos(2A + A)$

$$= \cos 2A \cos A - \sin 2A \sin A$$

$$= (2\cos^2 A - 1)\cos A - 2\cos A \sin A \sin A$$

$$= 2\cos^3 A - \cos A - 2\cos A \sin^2 A$$

$$= 2\cos^3 A - \cos A - 2\cos A (1 - \cos^2 A)$$

$$\boxed{\cos 3A = 4\cos^3 A - 3\cos A}$$

ii) Sub $x = 2\sqrt{2}\cos A$ into $x^3 - 6x + 2 = 0$

$$16\sqrt{2}\cos^3 A - 12\sqrt{2}\cos A + 2 = 0$$

$$16\cos^3 A - 12\cos A = -\frac{2}{\sqrt{2}}$$

$$4\cos^3 A - 3\cos A = -\frac{1}{2\sqrt{2}} \quad (\text{from previous})$$

$$\cos 3A = -\frac{1}{2\sqrt{2}} \quad (\text{from i})$$

$$\frac{-\frac{1}{2\sqrt{2}}}{\cos 3A} = -\frac{1}{2\sqrt{2}} \quad (\text{from qv})$$

$\therefore x = 2\sqrt{2}\cos A$ is a root of the equation $x^3 - 6x + 2 = 0$

iii) $\cos 3A = -\frac{1}{2\sqrt{2}}$

$$\therefore 3A = \pm \cos^{-1}\left(-\frac{1}{2\sqrt{2}}\right) + 2n\pi, \text{ where } n \text{ is a +ve integer}$$

Using $n = 1, 2, 3$ + +ve values of $\cos^{-1}$.

$$3A = 1.93216, \quad 8.21535, \quad 14.49853$$

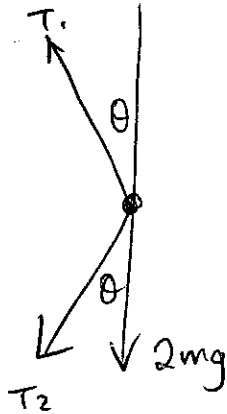
$$A = 0.64405, \quad 2.73845, \quad 4.83284$$

$$\therefore x = 2\sqrt{2}\cos A$$

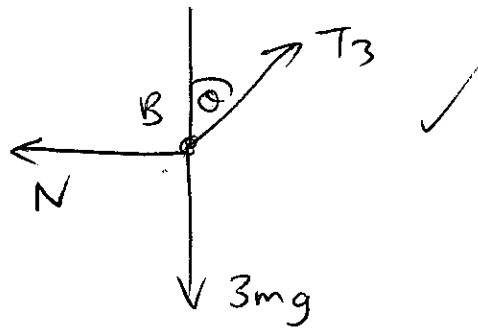
$$x = 2.262, \quad -2.602, \quad 0.340 \text{ to 3 dec place}$$

8 a)

Forces on A



Forces on B



i)

Sum Vertical forces

$$2mg + T_2 \cos \theta = T_1 \cos \theta \quad - (1)$$

Sum Horizontal forces

$$T_2 \sin \theta + T_1 \sin \theta = 2mr\omega^2 \quad - (2)$$

$$T_2 = T_3$$

$$\therefore T_1 = T_2 + \frac{2mg}{\cos \theta} \quad (- \text{from } (1))$$

$$T_1 = \frac{3mg}{\cos \theta} + \frac{2mg}{\cos \theta} \quad (\text{from } (3))$$

$$\boxed{T_1 = 10mg}$$

$$\boxed{T_2 = 6mg} \quad (\text{from } (3))$$

$$ii) \quad T_1 \sin \theta + T_2 \sin \theta = 2mr\omega^2$$

$$\text{sub } r = l \sin \theta$$

$$T_1 \sin \theta + T_2 \sin \theta = 2ml\omega^2$$

$$\omega^2 = \frac{T_1 + T_2}{2ml}$$

$$\omega^2 = \frac{16mg}{2ml} \quad \therefore \omega = \sqrt{\frac{8g}{l}}$$

Sum vert. forces

$$T_3 \cos \theta = 3mg \quad - (3)$$

Sum horizontal forces

$$T_3 \sin \theta = N \quad - (4)$$

(5)

iii) height above centre of circle decreases, or radius of circle increases.

(1)

(2)

Q8 b)

In order to prove L, M and N are collinear, we can show $\angle LMA = \angle NMB$

Step 1

In Δ 's PKM & BKN

$$\angle BKN = \angle PKM \text{ (vert opp.)}$$

$$\angle BKN = \angle PMK \text{ (90° given)}$$

$$\therefore \Delta PKM \parallel \Delta BKN \text{ (AAA)}$$

$$\therefore \frac{BK}{PK} = \frac{NK}{MK}$$

In Δ 's PKB & MKN

$$\angle PKB = \angle MKN$$

$$\frac{BK}{NK} = \frac{MK}{PK}$$

$$\therefore \Delta PKB \parallel \Delta MKN \text{ (2 sides ratio + included $\angle$)}$$

$$\therefore \angle NMB = \angle BPN \text{ (corr. $\angle$'s $\parallel$ Δ 's)}$$

Step 2 PACB is a cyclic quad

$$\angle PAC + \angle PBC = 180^\circ \text{ (opp. $\angle$'s cyclic quad supp.)}$$

$$\text{and } \angle PBC = \angle PAL \text{ (ext. $\angle$ cyclic quad = opp. interior $\angle$)}$$

$$\therefore \Delta PNB \parallel \Delta PLA \text{ (AAA)}$$

$$\therefore \angle BPN = \angle APL \text{ (corr. $\angle$'s $\parallel$ Δ 's)}$$

Step 3

$$\Delta ALS \parallel \Delta PMS \text{ (AAA)}$$

$$\therefore \frac{PS}{AS} = \frac{MS}{LS}$$

$$\therefore \Delta MLS \parallel \Delta PAS \text{ (2 sides ratio + included $\angle$)}$$

$$\therefore \angle SPA = \angle LMA \text{ (corr. $\angle$'s $\parallel$ Δ 's)}$$

$$\therefore \angle LMA = \angle NMB$$

and $\therefore$ LMN are collinear.

6

